# Phase 3G | Pivoting undercarriage and turning

PULSIPHER FARMS / MOAPA VALLEY CORN MAZE

# Show how the coach turns

PF-TRAIN-S3G-R0 | 18 October 2026 UTC

**Preliminary geometric demonstration and engineering development. Not for fabrication, procurement release, vehicle testing or passenger operation.**

The owner correctly observed that the earlier installation viewer showed apparently fixed wheels. Those were straight-track cylinder placeholders. This issue adds two articulated bogie assemblies beneath the retained F1 coach frame, visible functional support relationships, exact left/right curve poses and the evidence required to turn the concept into real hardware.

| Completed in this issue | Still requires actual engineering evidence |
|---|---|
| Front and rear bogies pivot relative to the coach, with visible axles and functional bogie/support symbols. | Exact offered bogie, wheels, bearings, suspension, brakes, support stack, mass and ratings. |
| Straight and left/right constant-radius geometry; calculated yaw angles and geometric clearance screens. | Full suspension movement, transitions, reverse curves, tolerances, wear, forces and vehicle dynamics. |
| Four coordinated drawing sheets, editable calculations, CAD poses and interactive browser demonstration. | Material, joint, weld, bolt, spring, pin, bearing and manufacturing dimensions; acceptance criteria and signoff. |

**What the visible hardware means:** springs, axleboxes, bogie frames, support pads and pivot symbols explain functions. Their drawn dimensions are display choices, not selected products or fabrication specifications. Do not extract their sizes from the mesh or STEP files for manufacturing.

# 01 | Retained basis and model use

Published Phase 3F version 8 and its files remain available unchanged. G1 carries forward the named F1/E1 32-inch-floor integration geometry, including all 110 frame member envelopes. This does not close the floor-height decision or replace the earlier 28-inch project baseline. The Phase 3E larger movement-envelope concern remains open.

| Input | Current meaning |
|---|---|
| Vehicle configuration | 36-inch / 914.4 mm evaluation gauge; three regular coaches and one accessible coach; diesel locomotive. CR-G01 changes the current brake preference to compressed-air service and spring-applied, air-released parking with commonly sourced combination chambers. |
| Frame and bogie locations | 400-inch frame, 110-inch outside width; front bogie x45 and rear x261 inches; 216-inch spacing. Body centre x153. These are retained study dimensions. |
| Wheel comparison | 596 mm cylinder diameter, 1165 mm bogie wheelbase and 101.6 mm display width retained from the identified comparison. That source bogie is not selected or gauge-compatible. |
| Track and route | 0% longitudinal-grade target. Constant-radius cases are 30, 50 and 75 m plus straight. They are geometric comparisons, not approved route radii. |
| Direction convention | Forward travel is toward decreasing body x. Vehicle left is negative y and right is positive y. The labels follow that convention. |
| Weight and performance | Retained coach/train mass scenarios remain allocations. New display geometry does not establish a mass, load rating, speed limit or operating capability. |

Open the new turning viewer, choose a curve and use the top view to compare the two bogie angles. The coach is held fixed on screen; the assemblies and track references change around it. Orbiting the camera and pivoting a bogie are separate functions. Play varies curvature between geometric poses. It is not a train driving through a surveyed transition or a dynamic simulation.

The small plan view remains useful when WebGL is unavailable. Display controls affect visibility only. Hiding a frame member, floor panel or service box does not remove that physical obstruction from the engineering model.

# 02 | Turning geometry and clearance findings

**Preliminary geometric study, 18 October 2026. No running gear has been selected or approved.**

The coach uses two swivelling, four-wheel bogies. Each bogie carries two parallel wheelsets and rotates beneath the body; the wheelsets do not steer individually like a road vehicle. The previous static wheel cylinders did not communicate this movement. Phase 3G gives the drawings, functional model and viewer one shared, calculated turning definition. Symbolic support and suspension parts explain their functions; their shapes are not fabrication details.

The retained F1 body has a 32-inch study floor, 110 unchanged steel members and bogie centres at x=45 and 261 inches, 216 inches apart. The body centre is x=153. Wheelbase 1165 mm, wheel diameter 596 mm and wheel-centre planes at ±457.2 mm remain **geometric proxies**, not accepted components or wheel/rail dimensions. The 36-inch evaluation gauge remains 914.4 mm. The track grade is zero.

## Exact geometry and its limits

Forward travel is toward **−x**. Positive y is the vehicle's right side when facing forward. Therefore LEFT uses negative curvature and RIGHT positive curvature. On a left curve the front bogie rotates positively and the rear negatively; the body remains fixed in the study view.

Let R be the circle radius through the four **axle midpoints**, h half the bogie wheelbase and L half the bogie-centre spacing. Using consistent units:

$Rb = \sqrt{R^2 - h^2}$; $Rc = \sqrt{Rb^2 - L^2}$.

For curve sign s, the circle centre is $(153, s \cdot Rc)$ and each bogie's yaw is $\operatorname{atan2}(s \cdot (x-153), Rc)$. A point $(u,v,z)$ in that bogie's local coordinates transforms to $(x + u\cos\psi - v\sin\psi,\ u\sin\psi + v\cos\psi,\ z)$. Local z remains the rail-reference ordinate; it does not invent a physical pivot height.

The bogie-centre radius vector is perpendicular to the bogie's longitudinal direction. Its axle midpoints are ±h along that direction, so their squared radius is $Rb^2+h^2=R^2$. This is an exact rigid-body construction. It is not a rolling-contact or steering-force solution: each axle's rolling direction differs from the local track tangent by $\pm\operatorname{asin}(h/R)$. Actual wheel profiles, conicity, clearances and suspension govern how a real bogie negotiates that mismatch.

| Axle-midpoint radius | Bogie yaw magnitude | Axle attack-angle magnitude | Body centre inward offset | Bare-frame inside / outside reach from track centreline |
|---|---|---|---|---|
| Straight | 0° | 0° | 0 in | 55 / 55 in |
| 30 m | 5.24745° | 1.11256° | 5.17172 in | 60.17172 / 65.97030 in |
| 50 m | 3.14527° | 0.66751° | 3.09868 in | 58.09868 / 61.77619 in |
| 75 m | 2.09618° | 0.44500° | 2.06488 in | 57.06488 / 59.58181 in |

The 400-by-110-inch frame extends 92 inches beyond each nearest bogie pivot. With frame half-length 200 and half-width 55 inches, inside reach is $R - Rc + 55$; outside reach is $\sqrt{200^2 + (Rc+55)^2} - R$. These are bare-frame radial envelopes, excluding steps, guards, doors, couplings and suspension movement. They are not route clearances or minimum-radius approvals. The displayed rail pair is a guide; proxy wheel centres are not proven contact points on those rails.

## What supports the body while it swivels

Railteco's RTKZ1 description documents a centre plate sharing support with sprung side bearers, plus axlebox springs and damping. Its detail listing explains the inherited 1165/596 mm proxy, but the actual product is 762 mm gauge with a 3.75 t axle rating. It is unsuitable as supplied; these numbers do not constitute a project offer. Its radial-link mechanism also differs from this rigid-bogie model.

CFD's railcar bogie demonstrates separate primary suspension, secondary support and traction rods. Its locomotive bogie uses a hemispherical support plate and side plates. These establish possible architectures,

not matching parts. The vertical load path generally runs body supports → bogie frame → primary suspension → axleboxes/bearings → wheelsets → rails. Longitudinal force, lateral force, roll moment and lift retention require their own defined load paths. A visible pivot does not prove that a single pin can safely carry them all.

Schaeffler describes application-designed railway axleboxes; GHH-BONATRANS coordinates wheelsets with bogies, brakes and track. The actual support height, bearings, wheel profile, suspension, stops and attachments remain null. The retained bolster underside is 482.7 mm above rail; **that is a body datum, not an approved support height**. Sources and limitations are dated in turning_sources.json.

### Clearance result and next evidence

The portable calculation checks exact finite wheel cylinders against retained solid outer steel boxes at seven poses: straight and left/right 30, 50 and 75 m. Minimum steel gap is 89.900 mm at curved rest states and 64.500 mm with 25.4 mm relative upward wheel movement; straight values are 89.905 and 64.505 mm. This limited zero-roll proxy screen finds no overlap. It omits actual flanges, axleboxes, bogie frames, springs and brake hardware. The earlier E1 P3 inflated/rolled bounding-box screen remains **−24.825 mm possible overlap**; the present narrower check does not close that unresolved condition.

Owner change **CR-G01** now directs air service brakes and spring-applied, air-released parking, investigating semi-truck-style chambers. The frozen Phase 3F hydraulic equipment reservations are historical comparison boxes only. Their clearance results provide no evidence that the new air equipment fits or is suitable for rail service.

bogie_sources_g.json holds supplier returns GR01-GR12. turning_interfaces.json sequences exact configuration and support returns, load-path/brake design, full moving/removal clearance, agreed verification protocols and documented installation/testing. Actual evidence and acceptance fields remain blank. Python/JavaScript parity, exact-circle closure and left/right checks verify this mathematical model; they do not replace OEM application acceptance, professional review, testing or commissioning.

# 03 | How the coach is supported

The intended load path is from the coach structure through designed body-to-bogie supports, through the bogie structure and suspension, into axleboxes, bearings, axles, wheels and rail. The offered architecture determines the order and sharing of those functions. A centre pivot symbol identifies the rotational relationship; it does not imply that a kingpin carries the whole vertical load or that a drawn disc is a selected slewing bearing.

| Function | What must be resolved before release |
|---|---|
| Vertical support and roll control | Identify the actual centre/side supports, load sharing, loaded/unloaded heights, preload, wear, fatigue and body attachment reactions. |
| Yaw movement | Provide real rotation surfaces or bearings, yaw resistance, stops, retention and the permitted rotation with all brakes, hoses and services installed. |
| Suspension and wheel contact | Provide springs/dampers, bounce/rebound, roll/pitch/twist response, wheel profiles and rail/turnout compatibility; verify wheel unloading and the complete operating envelope. |
| Longitudinal and lateral forces | Resolve the braking, traction/draft, lateral and combined force paths without assuming that one illustrated connector carries every load. |
| Installation and retention | Issue mounting details, fits/tolerances, lubrication, assembly procedure, lifting/jacking points, retention, access and inspection criteria. |

Farm crew work can advance through ground-level templates and information gathering. Building bogie frames, machining wheelsets, welding support attachments or installing safety-critical running gear requires the responsible specialists and released drawings. No drilling or cutting of the retained frame is authorized by this phase.

# 04 | Air-brake change CR-G01

Owner direction received 18 October 2026: use air brakes with canisters providing parking-brake capacity and easy sourcing comparable to a semi-truck. This supersedes the hydraulic preference prospectively. Phase 3F and earlier reports remain preserved historical issues; their hydraulic module locations are not an air-brake installation design.

PF-TRAIN-S3G-R0 · 18 October 2026 UTC · **Owner-directed design change; no component selection or installation release.**

Develop compressed-air service braking with spring-applied, air-released parking brakes. Common truck combination chambers are a credible procurement direction. They supply actuator force, not a complete rail brake. Preserve Phase 3F as the superseded hydraulic baseline; its pump loads, pressure-release circuits and hydraulic installation boxes are no longer current air-system selections. Emergency lighting remains independent.

## Real chamber candidates

| Candidate for inquiry | Documented facts | What remains unproven |
|---|---|---|
| MGM TR3030, **3430051** | Standard 2.50 in stroke; size 30 service / 30 spring; 8 in rod without yoke. Current product page and 5019 brochure identify this exact variant. | Force versus pressure/stroke, minimum spring force, approved brake mechanism and rail application |
| MGM TR3030LP3HD, **3232051** | 3.00 in stroke; brochure mass 20.4 lb. HD brochure states approximately 145 lbf additional spring force at 1.25 in stroke versus its standard comparison. | Absolute force curve and tolerances; the increment is not a parking rating |
| Haldex LifeSeal+ **LCP3030HD** | 2.50 in stroke, 10 in threaded rod, heavy-duty housing and high-output spring in the manufacturer guide | Complete envelope, force curves, application approval and current configured availability |

These are alternative assemblies, not interchangeable selections. The MGM 3430051 brochure gives housing dimension A = 9.04 in, service diameter C = 8.32 in, spring diameter D = 8.21 in and mass 19.4 lb. Rod travel, fittings, caging access and replacement clearance add to that envelope. Its diameter already exceeds the old 7 in hydraulic reservation height: those boxes cannot be relabeled as chamber fits. [C3GA01-05]

Bendix currently lists EnduraSure and EnduraSure-Pro as another manufacturer route, but an exact part number and application data remain required. MGM publishes replacement parts and a distributor locator; its 3430051 page displayed a two-day lead time when checked. That is a public listing, not a quote, local stock confirmation or commitment to this project. Obtain dated distributor returns and approve one controlled spares configuration after engineering. [C3GA01, C3GA07]

## Match force to the rail brake

A “30/30” designation is not 30 kN, a vehicle capacity or a guaranteed parking force. The nominal size-30 service area gives a useful ideal pressure-times-area screen; spring force and actual service output require model-specific stroke curves. At an illustrative 60 psi, 30 in² produces 1,800 lbf before return-spring, friction and effective-area corrections. This is **not** a selected operating pressure or an actuator rating. [C3GA08]

The retained 13,200 kg regular-coach case needs 6.60 kN at 0.50 m/s² or 10.56 kN at 0.80 m/s² on level track, before additional dynamics. With four equally braked axles and the 298 mm radius proxy, the latter is 786.72 N·m per axle. Applying that same force to the 7,040 kg tare case would demand 1.50 m/s² and ideal adhesion of 0.153. Empty/load balance therefore matters more than buying the largest available chamber.

The calculation register leaves actuator count, brake torque per unit pushrod force, service pressure, allowable stroke, spring-force curve and parking demand blank. An S-cam chamber and slack adjuster do not by themselves define wheel torque. Select a matched rail disc, tread or drum foundation brake, then verify rigging travel, mechanical advantage, wear adjustment, heat, mount reactions and wheel adhesion. Parking must hold after leakage and loss of power; zero design grade does not eliminate wind, residual slope, disturbances or recovery restraint.

## Proposed whole-train functions

The configuration inquiry should cover a locomotive compressor, governor/control, dryer, protected supply reservoir, relief/drains and pressure indication; a continuous protected supply path; and local coach reservoirs/check valves and relay functions. Keep service demand distinct from the automatic emergency/separation function. Evaluate an automatic rail brake-pipe arrangement against a properly engineered supply/control trainline; do not copy one semi-trailer schematic across four coaches.

Each car must apply a defined emergency response after separation, low pressure or control failure, without waiting for hotel electricity. Preserved reservoir pressure must not trap the spring brakes released after separation. Conversely, a check valve or reservoir fault must not cause an uncontrolled isolated application. Assess supply-line rupture, service-line rupture, stuck relay, blocked exhaust, frozen condensate, crossed connections and a disconnected car whose hoses remain attached. Valve thresholds, air capacities, compressor duty, line sizes and response times remain held.

Anti-compounding prevents service and spring forces from simply adding and overloading the mechanism. Haldex documents FF2 variants with this feature and separate control, supply, service-delivery and spring-delivery connections. It is a functional reference, not a train valve selection. Validate overlap and transition behavior so avoiding overload never creates a brake-release interval. Road-vehicle regulatory references in manufacturer manuals do not establish this attraction’s jurisdiction. [C3GA06, C3GA08]

## Required returns and commissioning

Require exact chamber drawings; pressure/force/stroke curves; spring minimum-force and release-pressure limits; full worn-brake travel; permitted orientation; mounting reactions; hose motion; environmental ratings; caging access; and approved substitution/spares rules. The brake integrator must return the complete pneumatic schematic, reservoir/compressor calculations, failure analysis and empty/full stopping and parking criteria before procurement release.

Instrument pressure at the locomotive, each local reservoir and representative service/spring ports; measure pushrod travel, wheel speed and synchronized stopping response. Reuse the Phase 3E commissioning framework with revised air-specific procedures: charge/recharge, leakage, low-pressure warning, controlled separation, full-train timing, anti-compounding, empty/full balance, parking and recovery restoration. Limits and test loads require OEM/engineer acceptance before testing.

Caging mechanically removes spring parking action; it belongs to restrained, unoccupied recovery under the exact manufacturer procedure. Factory-sealed spring chambers must not be opened or rebuilt by the farm crew. Qualified brake technicians should install and commission the matched system; the crew can prepare approved routes, labels and inspection access. [C3GA04]

# 05 | Actual bogie selection and closure records

PF-TRAIN-S3G-R0 · 18 October 2026 UTC · **Inquiry candidates only. No selected bogie, quotation, availability commitment or supplier acceptance. No contact made.**

Develop one complete braked passenger-bogie specification with a named design authority. CR-G01 changes the prospective basis to air service braking with spring-applied, air-released parking, including investigation of readily sourced truck-style chambers. This does not select a chamber, foundation brake or approved rail application. These three manufacturers have relevant primary evidence; none of the retrieved material establishes an off-the-shelf assembly for this coach. Purchasing a nominal 36-inch wheelset separately would leave the body support, suspension, braking and wheel/rail interface unresolved.

| Candidate | Verified relevance | What the evidence does not establish |
|---|---|---|
| **Severn Lamb - complete-system or engineered undercarriage inquiry** | Its JUPITER page identifies nominal 914 mm gauge, enclosed-coach availability and a 50 m minimum radius. Its custom page covers gauges from 15 inches through standard gauge. [G-B01-02] | No offered bogie-only model, support height, axle load, wheelbase or project-specific radius. JUPITER's radius belongs to that complete product; it cannot be transferred to this longer coach or to an unnamed bogie. |
| **Alan Keef Ltd - bespoke bogie/body-interface inquiry** | The manufacturer documents a pair of new Ashbury-carriage bogies with roller-bearing axleboxes. It also documents weather-protected, heated passenger coaches with an accessible carriage. [G-B03-04] | No retrieved 36-inch project offer or rated bogie matching the current mass and floor. Smaller-gauge passenger references demonstrate relevant experience, not scalable ratings. |
| **Western Train Co. - regional complete-coach/custom-parts inquiry** | Its FAQ offers custom gauge work and lists 36 inches among train gauges. The manufacturer supplies air-braked passenger coaches, including an enclosed cabin-coach reference. [G-B05-06] | No controlled pivot-bogie drawing, support stack, rating or radius for this car. Its air-brake architecture aligns at a high level with CR-G01; no supplied spring-parking or complete train-brake configuration is established. Bogie-only supply is unconfirmed. |

This ordering is an inquiry strategy, not a vendor ranking by cost or certified capability. Severn Lamb offers the clearest published same-nominal-gauge complete-system reference; Alan Keef provides direct evidence of separate bogie manufacture; Western Train provides a relevant domestic integration route. A manufacturer may reasonably require its own engineered coach or reject the proposed floor and mass. Preserve that option instead of forcing a purchase to match the drawing.

A 36-inch project gauge is exactly 914.4 mm by conversion. A marketing entry of 914 mm must lead to a controlled nominal-gauge/tolerance return, not an assumed 0.4 mm manufacturing discrepancy. Gauge alone does not specify wheel back-to-back, profile, tread contact, flange clearance, rail cant or turnout compatibility.

## Submit one coordinated interface basis

The retained study body has bogie centres at x = 45 and 261 inches, 216 inches apart, and a 400-inch frame. F1 retains the 32-inch floor study and a nominal bolster underside at 482.7 mm above rail. These are packaging constraints to test, not an accepted body-bearing elevation. The 13,200 kg regular-coach upper mass case and the earlier 8,295 kg maximum longitudinal bogie-plane rail reaction are analytical inputs; neither is a body-support load rating. Bogie self-weight and suspension/load sharing must be resolved before deriving centre-bearing and side-bearing forces.

Request an identifiable model/configuration and controlled drawings for both bogies, wheelsets, brake assemblies and the vehicle connection. An effective return should state:

1. **Support and load paths:** loaded/unloaded support heights; centre-bearing, side-bearing and yaw-guide coordinates; which parts carry vertical load, longitudinal braking/draft load, lateral load, roll moment and retention. Do not assume a vertical kingpin carries the entire body weight or that an illustrated disc is a purchased slewing bearing.

2. **Load case acceptance:** sprung/unsprung mass split, axle/wheel loads, empty/full and eccentric occupancy, body CG limits, curve/twist/dynamic cases, brake force/torque paths and fatigue basis. The load spectrum and accepted limits require the vehicle engineer and supplier together.

3. **Movement and limits:** full yaw, roll, pitch, bounce/rebound and wear envelopes; spring/damper characteristics; stops and retention; wheel, axlebox, brake and hose volumes; curve/reverse-curve limits for the actual body and coupling geometry. The Phase 3G exact-circle animation is a planar kinematic study, not a minimum-radius approval.

4. **Wheel/rail compatibility:** controlled wheel/axle drawing, actual rolling/new/worn diameters, back-to-back, tread/flange profile and bearing arrangement, tied to the track engineer's rail, fastening, gauge and turnout design.

5. **Installation and service:** actual mass/CG, mounting loads and permitted attachment details, tolerances, assembly/inspection methods, brake replacement access, jacking/retention requirements and controlled maintenance documentation. No frame drilling, weld, pin size or spring selection follows from the visual support proxies.

The JSON provides twelve concise return records with actual values and acceptance left null. It also explicitly leaves every candidate's project support dimensions, ratings, wheelbase, radius, price and lead time null. A published brochure is a research record, not a procurement response.

### Next useful release

After authorization to contact suppliers, send the same configuration basis and return schedule to the chosen candidates. Review a controlled CAD/data return against the untouched frame, wheels and services; then choose the floor/support arrangement jointly with the body and boarding design. Release the actual interface analysis, matched air-brake integration and unoccupied fit/rotation trial before any occupied test. Retain the Phase 3E qualification order - component evidence, inspected assembly, instrumented unoccupied verification and only then approved supervised participant work - while revising equipment-specific brake procedures and criteria for CR-G01. This memo authorizes no fabrication or testing.

# 06 | Focused qualification sequence

| Gate | Evidence and responsible completion |
|---|---|
| 1. Offered configuration | Bogie supplier and vehicle engineer return controlled part/configuration identity, CAD, wheel/rail interfaces, brake architecture, loads and motion limits. |
| 2. Coordinated installation | Vehicle/structural, brake and track engineers resolve support height, frame connections, floor/boarding consequences, service routing and complete clearances. |
| 3. Controlled unoccupied fit trial | Qualified team assembles the released configuration, records actual heights/masses and checks permitted yaw, suspension movement, stops, retention, hoses and maintenance access under an approved procedure. |
| 4. Loaded qualification | Use engineered test loads, accepted instruments and criteria to verify the wheel-load, stability, braking and curve/transition cases. Resolve nonconformances before duplication or participant work. |

The workbook provides editable geometric calculations and blank supplier/verification records. Actual measured results, inspection acceptance and approval fields remain blank. Digital export checks establish file and calculation consistency only. They do not qualify a physical vehicle.

Formats: readable PDF sheets; vector SVG; declared paper-inch and actual-inch DXF geometry; STEP in millimetres; web geometry in inches; editable XLSX and portable calculation/model scripts. Functional proxy geometry is explicitly marked in the model metadata. Native engineering CAD assembly constraints and manufacturing features are not supplied.

# 07 | Source register and verification scope

Primary sources were reviewed on 18 October 2026 UTC. Catalog evidence is not a supplier acceptance, quotation, stock check or project approval. Earlier study inputs are retained with their limits. No supplier was contacted through this phase.

## TG-S01 | RTKZ1 passenger bogie corporate product description

Document date: Undated public product page | Accessed: 18 October 2026 UTC

762 mm passenger bogie; centre plate and sprung side bearers share body support; primary axlebox springs and hydraulic damping; radial-link arrangement; 596 mm wheels.

Limit: Different gauge and radial-link architecture. Neither the support geometry nor this manufacturer's curving behavior is represented by the project's rigid-bogie proxy.

https://cn.railteco.com/cplist.aspx?cptp=8&i=14

## TG-S02 | RTKZ1 passenger bogie product detail

Document date: Undated public product page | Accessed: 18 October 2026 UTC

Body of listing identifies RTKZ1: 762 mm gauge, 1165 mm wheelbase, 596 mm wheels, 3.75 t maximum axle load, bogie tare no more than 1.5 t and maximum speed 40 km/h.

Limit: Rejected as supplied: 762 mm is not 914.4 mm, and two 3.75 t axle ratings do not cover the inherited 8.295 t maximum bogie-plane rail-reaction sensitivity. Page heading has copy inconsistencies; corporate RTKZ1 description corroborates identity. No accepted project offer, drawing or rating.

https://www.railwayspare-parts.com/sale-13771970-aar-standard-762m-gauge-passenger-train-bogie.html

## TG-S03 | Railcar bogie

Document date: Undated public product page | Accessed: 18 October 2026 UTC

Two-stage design with primary axlebox coils and dampers, secondary air suspension, traction rods linking body and bogie, and dampers controlling body motions. Published 2.4 m wheelbase and gauges 1000, optional 1067, and 1435 mm.

Limit: Architecture example only: not the proposed gauge or dimensional package; air suspension is not selected or required here.

https://www.cfd.group/rolling-stock/railcar-bogie

## TG-S04 | BB locomotive bogie

Document date: Undated public product page | Accessed: 18 October 2026 UTC

Primary coil suspension; hemispherical support plate and two side plates connect bogie and locomotive frame. Published 2.5 m wheelbase and 1000/1435 mm gauges.

Limit: Locomotive architecture example only, not an accepted coach part or support-height reference.

https://www.cfd.group/rolling-stock/lomocotive-bogie

## TG-S05 | Axlebox bearings for passenger cars and locomotives

Document date: Undated public technical page | Accessed: 18 October 2026 UTC

Railway axlebox systems combine application-designed housings and bearings; the maker identifies cylindrical and tapered bearing families with application-specific seals and lubrication.

Limit: No bearing size, fit, load rating, life or maintenance interval has been selected for this project.

https://www.schaeffler.us/us/products-and-solutions/industrial/industry_solutions/rail/axlebox_bearings_passenger_cars_locomotives/

## TG-S06 | Wheelsets

Document date: Undated public product page | Accessed: 18 October 2026 UTC

Manufacturer describes wheelsets developed in coordination with bogies, brakes, drives and track parameters, across gauge types.

Limit: No actual wheel profile, back-to-back, axle geometry or accepted component rating is returned.

https://www.ghh-radsatz.com/en/products-and-services/wheelset/

## G-B01 | JUPITER

Document date: Undated public page | Accessed: 18 October 2026 UTC

Nominal 914 mm complete passenger-train reference, enclosed option and 50 m product minimum radius.

Limit: No separate bogie rating or transferable project radius.

https://www.severn-lamb.com/people-movers/tracked-trains/rail-trains/jupiter-2

## G-B02 | Custom trains

Document date: Undated public page | Accessed: 18 October 2026 UTC

Custom gauge range 15 inches to 4 ft 8.5 inches.

Limit: Gauge range does not establish custom offer, rating or availability.

https://www.severn-lamb.com/people-movers/tracked-trains/custom

## G-B03 | Heritage gallery: Ashbury carriage bogies

Document date: Undated public page | Accessed: 18 October 2026 UTC

Manufactured new carriage bogies with roller-bearing axleboxes.

Limit: No retrieved 36-inch configuration, support dimensions or rated load.

https://www.alankeef.co.uk/gallery/heritage-restoration

## G-B04 | Leisure, park and farm gallery

Document date: Undated public page | Accessed: 18 October 2026 UTC

Weather-protected heated coaches with accessible carriage; relevant passenger integration experience.

Limit: No project-specific bogie acceptance or scaling of another gauge.

https://www.alankeef.co.uk/gallery/leisure-park-farm

## G-B05 | FAQ

Document date: Undated public page | Accessed: 18 October 2026 UTC

Custom gauge work, 36-inch train listing, custom parts and air-braked passenger coaches.

Limit: The FAQ uses wheelbase in one gauge answer; controlled gauge and actual axle spacing must be returned separately. No bogie-only offer.

https://www.westerntrain.com/faq

## G-B06 | Coach and train product page

Document date: Undated public page | Accessed: 18 October 2026 UTC

Passenger and enclosed cabin-coach references with air brakes.

Limit: No support geometry, axle load or minimum radius for proposed car; photographs do not disclose hidden engineering.

https://www.westerntrain.com/locomotives

## C3GA01 | 3430051 / TR3030 current product listing

Document date: Undated live product page | Accessed: 2026-10-18

Exact 3430051: 2.50 in stroke; service/spring sizes 30/30; 8.00 in rod without yoke; replacement parts listed. Public lead time displayed 2 days.

Limit: No quote, local inventory confirmation, project acceptance, absolute force curve or load rating.

https://www.mgmbrakes.com/products/3430051

## C3GA02 | Form 5019 TR Model Spring Brakes

Document date: Rev. 5-2018; currently linked by manufacturer | Accessed: 2026-10-18

3430051: mass 19.4 lb; A housing length 9.04 in; C service diameter 8.32 in; D spring diameter 8.21 in. Page 2 figure and table inspected.

Limit: Not a complete installation envelope: protruding fittings, rod stroke, caging and removal clearance remain additional.

https://www.mgmbrakes.com/download_file/8ffc37a4-da80-4e10-85cb-417ecbec5a08/9

## C3GA03 | Form 5033 Heavy-Duty Spring Brakes

Document date: Rev. 2-2021; currently linked by manufacturer | Accessed: 2026-10-18

3232051 TR3030LP3HD: 3.00 in stroke and 20.4 lb. Brochure states approximately 145 lbf additional spring force at 1.25 in stroke in its comparison.

Limit: The force increment is not an absolute guaranteed curve or holding capacity. Do not derive minimum spring force from the graph.

https://www.mgmbrakes.com/download_file/0d900468-f876-4535-be3f-52af16266849/9

## C3GA04 | S-Cam Drum & Disc Actuator Service Manual EB 24-001

Document date: Rev. 02, November 2024 | Accessed: 2026-10-18

Specifies exact actuator/stroke identification, restrained maintenance, caging procedures and factory-sealed spring chamber restrictions.

Limit: Selected chamber and foundation-brake instructions still govern. No project installation torque, pressure or procedure released here.

https://www.mgmbrakes.com/download_file/7c0e38b7-5b17-4cd2-b065-bbb9b8f7a16d/9

## C3GA05 | Actuator Quick Reference Guide

Document date: Document footer L20751, US Rev. 3/19; currently hosted | Accessed: 2026-10-18

LCP3030HD has 2.50 in stroke, 10.00 in threaded pushrod, heavy-duty housing and high-output spring. LCP3030LHD is a distinct 3.00 in stroke option.

Limit: No absolute force curve or full envelope; current configured availability requires a return. Document number in footer differs from URL filename.

https://www.haldex.com/globalassets/north-america/documents/actuators/l20600

## C3GA06 | L31107 FF2 Full Function Valve installation instructions

Document date: Rev. 2/02; currently hosted | Accessed: 2026-10-18

Identifies supply, control, service-delivery and spring-delivery functions and FF2 anti-compounding variants.

Limit: Older road-trailer reference only. No valve selection, trainline timing, project pressure or attraction-code applicability inferred.

https://www.haldex.com/globalassets/north-america/documents/valves/l31107

## C3GA07 | Actuators: EnduraSure Spring Brake Series

Document date: Undated live manufacturer page | Accessed: 2026-10-18

Current EnduraSure and EnduraSure-Pro product families, technical-documentation and authorized sales channels are listed.

Limit: No exact part, curve, dimensions, stock or project application confirmed. Marketing durability comparisons not used for design.

https://www.bendix.com/en/products/wheel-end-solutions/actuators/

## C3GA08 | Air Brake Handbook BW5057P6

Document date: Historical manufacturer handbook in agency archive; publication date not established | Accessed: 2026-10-18

Explains type-30 nominal service area of 30 $in^2$, local relay functions and anti-compounding. Explicitly identifies its schematics as illustrative rather than actual vehicle piping.

Limit: Conceptual corroboration only, not a current parts manual or applicable rail regulation. Current configured OEM returns are required.

https://static.nhtsa.gov/odi/inv/2020/INRD-PE20003-10834.pdf

Verification records accompany the source package: canonical curve calculations, independent model/geometry checks, document visual review, workbook recalculation/export checks and publication checks. Their recorded limits apply. Browser execution is reported separately from static code and geometry checks; no physical prototype test or engineering approval is claimed.